

Answer ALL questions.

Write your answers in the spaces provided.

Some questions must be answered with a cross in a box ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

- 1** The cell membrane is important in the control of which substances can enter and leave the cell.

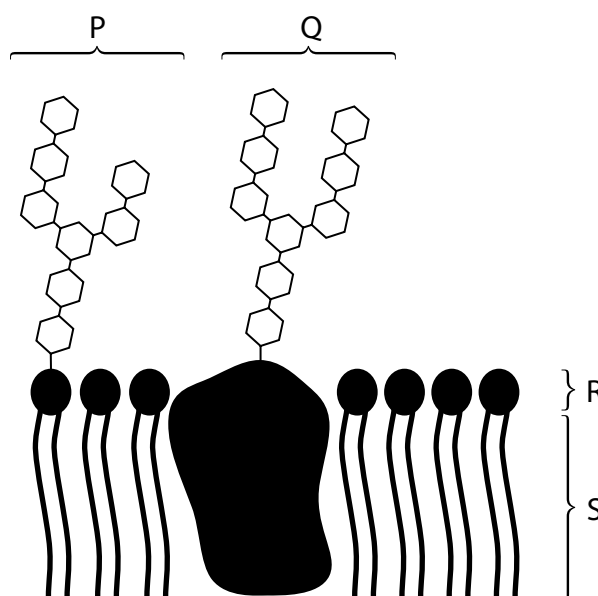
(a) The cell membrane consists of a phospholipid bilayer.

(i) Why do phospholipids form a bilayer?

(1)

- ☐ **A** the hydrophobic heads dissolve in the aqueous (water) environment
- ☐ **B** the hydrophobic heads move away from the aqueous (water) environment
- ☐ **C** the hydrophobic tails dissolve in the aqueous (water) environment
- ☐ **D** the hydrophobic tails move away from the aqueous (water) environment

(ii) This diagram shows part of a cell membrane.



Which letter represents a membrane glycoprotein?

(1)

- ☐ **A** P
- ☐ **B** Q
- ☐ **C** R
- ☐ **D** S

DO NOT WRITE IN THIS AREA

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(b) State what is meant by the term **osmosis**.

(1)

(c) Compare and contrast exocytosis and endocytosis.

(3)

(d) Explain why oxygen molecules can pass directly through the cell membrane.

(3)

(Total for Question 1 = 9 marks)